

Prayas 2.0 2027

Chemistry (GOAT)

Some Basic Concepts of Chemistry

DPP: 10

Total Duration - 30 Min

- Q1** CaCO_3 is 90% pure. Volume of CO_2 collected at STP when 10 g of CaCO_3 is decomposed is-
 (A) 2.016 litres
 (B) 1.008 litres
 (C) 10.08 litres
 (D) 20.16 litres
- Q2** 33.6 g of an impure sample of sodium bicarbonate when heated strongly gave 4.4 g of CO_2 . The percentage purity of NaHCO_3 would be:
 (A) 25% (B) 50%
 (C) 75% (D) 100%
- Q3** 20g of an impure sample of calcium carbonate decomposes on heating to give 8.4g of calcium oxide. What is the percentage purity of calcium carbonate sample?
 [Atomic weight of Ca is 40u]
 (A) 85 (B) 50
 (C) 95 (D) 75
- Q4** 23 g of ethanol ($\text{C}_2\text{H}_5\text{OH}$) on reaction with ethanoic acid (CH_3COOH) forms 44 g of ethyl ethanoate ($\text{CH}_3\text{COOC}_2\text{H}_5$) and some weight of water. The percentage yield of the reaction is
 (A) 50% (B) 100%
 (C) 25% (D) 12.5%
- Q5** In the following reaction,
 $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$
 2 moles of MnO_2 react with 4 moles of HCl to form 11.2L of Cl_2 at STP.
 Thus, percent yield of Cl_2 is
 (A) 25% (B) 50%
 (C) 100% (D) 75%
- Q6** The right option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is
 (Atomic mass of Ca = 40)

$$\left[\text{CaCO}_3 \xrightarrow{1200\text{ K}} \text{CaO} + \text{CO}_2 \right]$$

 (A) 1.76 g (B) 2.64 g
 (C) 1.32 g (D) 1.12 g
- Q7** What is the meaning of 10% solution of NaCl?
 (A) 10 g NaCl in 100 gram of solution
 (B) 58.5 grams of NaCl in 100 grams of solution
 (C) 94 grams of NaCl in 100 grams of solution
 (D) 5.85 grams of H_2O in 100 grams of solution
- Q8** A solution is prepared by adding 5 g of a substance x to 18 g of water. Calculate the mass percentage of the solute.
 (A) 21.74% (B) 19%
 (C) 25% (D) 40%
- Q9** Given, 5×10^{-3} kg of urea dissolved in 2×10^{-3} kg of water. Calculate the percent by mass of urea.
 (A) 90% (B) 71.42%
 (C) 70% (D) 80%
- Q10** Determine the volume/volume percent solution made by combining 25 mL of ethanol with enough water to produce 200 mL of the solution:
 (A) 12.5 (B) 20
 (C) 40 (D) 25
- Q11** Calculate the volume by volume percentage of a solution of 15 mL of alcohol in 60 mL of water.
 (A) 20 (B) 25
 (C) 30 (D) 50



Q12 Rashida dissolved 40 g of sugar in 600 mL of sugar solution. Calculate the mass by volume percentage.

- (A) 66.6 (B) 70
(C) 6.66 (D) 50



Answer Key

Q1 (A)

Q2 (B)

Q3 (D)

Q4 (B)

Q5 (B)

Q6 (A)

Q7 (A)

Q8 (A)

Q9 (B)

Q10 (A)

Q11 (A)

Q12 (C)



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